

Any dynamic burst in resource requirements can then be controlled within the allocated budget with the help of alerts and notifications.

- **Choose the right compute services for better performance and costs:** Though cloud service providers like Azure, AWS and GCP have multiple compute models and instance types, in many situations we may not know if the right compute size and model is being used. If we optimise this model taking into account performance and technical requirements, we can save costs.

Cloud service providers have internal financial governance defined as part of their cloud service expense management (CSEM), which helps in better cost management and transparency. Cloud architects and consultants must enable this while designing cloud adoption.

FinOps adoption helps new models in cloud management by bringing cloud engineers and finance teams together in an agile POD structure. Here, procurement is transparent and efficient through a defined workflow, which enables predictable costs and reasonable budgets.

FinOps also helps to address the cloud demand trade-offs like quality of service, speed in procurement and cost management. According to the FinOps Foundation, the key principles for FinOps adoption are:

- Create collaborative teams covering finance, architects, business sponsors and operations teams.
- Decisions for cloud adoption should be driven by business value.
- Cloud usage is not a single person or team's responsibility but is everyone's responsibility.
- Cost transparency is very important right from the early stage of adoption, and FinOps reports should be available to all stakeholders.
- A centralised FinOps team should be set up to define the cloud adoption roadmap and the policies.

- The FinOps framework should use the variable cost model from the cloud service provider (CSP).

Cloud cost management tools

The most effective and popular cloud cost management tools are listed below.

Apache CloudStack: Apache CloudStack is an IaaS cloud computing platform that helps businesses deploy virtual machines and manage geographically distributed data centres on a centralised server. IT professionals can customise the user interface and configure installed applications with features like firewalling, virtual machine (VM) templates, routing, storage replication, dynamic host configuration protocol (DHCP), and more. Administrators can use Apache CloudStack to manage access permissions, change account passwords and allocate resources to specific domains or users.

With its project management module, enterprises can send project invites to employees and organise users into multiple teams, enabling them to collaborate and share virtual resources such as snapshots, templates, data disks and IP addresses. Managers can verify and authenticate end users via the external Lightweight Directory Access Protocol (LDAP) servers such as Microsoft Active Directory and ApacheDS. Apache

CloudStack allows businesses to create new virtual machines and assign them to specific hosts or affinity groups based on individual requirements. Developers can create new network offerings with details, including name, description, data transfer rate, guest type, and virtual private cloud (VPC).

OpenStack: The OpenStack cloud cost management tool includes software tools that help in creating and managing cloud computing services, using pooled virtual resources. It contains 'projects', which are tools made using the OpenStack platform. These help to manage core cloud services such as computing, storage, networking, identity, and images. OpenStack tools can help to control large batches of computing, storage and networking, using resources from the data centre. These tools are managed by the OpenStack API or a dashboard.

ManageIQ: ManageIQ is another top open source cloud cost management tool. It offers help with managing small and large virtual environments and also supports advanced technologies, such as public clouds, virtual machines and containers. With ManageIQ users can download any virtual appliance and deploy its copies in virtualisation platforms like VMware or OpenStack. 

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